

Hammad Rahman

Senior Data Engineer

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PROFESSIONAL SUMMARY

Senior Data Engineer with **8+ years of experience designing large-scale data platforms and lakehouse architectures** across fintech and healthcare domains. Expert in **Apache Spark, Kafka, Databricks, and cloud-native data engineering on AWS and Azure**. Proven track record building real-time and batch pipelines processing **hundreds of millions of records monthly**, implementing **data governance and security frameworks**, and enabling **BI, machine learning, and advanced analytics at scale**.

CORE SKILLS

Data Engineering:

Apache Spark, PySpark, Spark Structured Streaming, Delta Lake, Databricks, ETL / ELT Pipelines

Streaming & Messaging:

Apache Kafka, Real-time Data Processing, CDC Pipelines

Cloud Platforms:

AWS (S3, Glue, Redshift, EMR, Lambda), Azure (Azure Databricks, Azure Data Factory, ADLS Gen2), GCP BigQuery

Data Orchestration & DevOps:

Apache Airflow, Terraform, CI/CD, GitHub Actions, Jenkins

Programming Languages:

Python, SQL, Scala, Java, Bash

Data Modeling & Governance:

Star Schema, Data Vault, Lakehouse Architecture, RBAC, Data Lineage, Metadata Management, PII Masking

Analytics & ML:

MLflow, Spark MLlib, Scikit-learn, TensorFlow

Data Visualization:

Power BI, Tableau

PROFESSIONAL EXPERIENCE

Senior Data Engineer

Chime Financial

09/2023 – Present

Remote

- Designed a **scalable lakehouse data platform** supporting analytics, **BI dashboards**, and **ML workloads** across multiple teams.
- Architected and maintained a **cloud-native lakehouse platform** supporting large-scale financial and telemetry datasets.
- Developed Spark and Kafka pipelines processing **200M+ records monthly**, enabling near real-time analytics and reducing reporting latency by **40%**.
- Implemented **enterprise data governance** with **Unity Catalog (RBAC, lineage, PII masking)**, improving **data access security** and audit readiness across 20+ datasets.
- Automated **data ingestion, validation, and transformation workflows** using Apache Airflow and Azure Data Factory.
- Enforced **HIPAA-compliant security controls**, including encryption, audit logging, and fine-grained access policies across AWS and Azure environments.

- Optimized storage and compute across **AWS and Azure environments**, reducing data platform costs by 20%.

Tech Stack: Spark, Kafka, Databricks, Delta Lake, Airflow, Azure Data Factory, AWS, Python, SQL

Big Data Engineer

Oscar Health

07/2020 – 08/2023

Remote

- Led development of **large-scale data platforms for healthcare and fintech clients** using Spark, Databricks, Delta Lake, and Kafka.
- Migrated 30+ legacy **Hadoop-based pipelines to modern lakehouse architectures**, reducing processing time by 35%.
- Built **ETL and ELT pipelines with PySpark, Airflow, and AWS Glue**, processing billions of records monthly.
- Implemented **CDC-based real-time ingestion pipelines** with Kafka and Delta Lake supporting exactly-once data delivery.
- Engineered **dimensional and data vault models** for enterprise analytics, reporting, and regulatory compliance.
- Established **CI/CD pipelines** using **Terraform, GitHub Actions, and Jenkins**, reducing deployment time by 50%.

Tech Stack: Spark, Kafka, Databricks, AWS Glue, Airflow, Terraform, Python, SQL

Data Engineer

InvoZone

10/2017 – 06/2020

- Developed **Spark-based data pipelines** processing 50M+ financial transactions monthly for analytics and fraud detection.
- Built optimized **warehouse data models powering real-time dashboards and operational analytics**.
- Delivered **clean, reliable datasets supporting risk analytics and fraud detection models**.
- Collaborated with analytics and product teams to translate business requirements into **production-grade data pipelines and datasets**.
- Improved **data pipeline performance** by 30% through query optimization, partitioning, and indexing strategies.

Tech Stack: Spark, Kafka, Python, SQL, Data Warehousing

EDUCATION

Bachelor of Science in Computer Science

COMSATS University